

OVERLIFT NES

Master Engineering Handoff

Completed iterations, current branch state, cartridge test library, and the compatibility waves ahead

Current canonical target

Iteration 02B.4A / 02B.4B

Unify the three strongest source branches, integrate every valid supplied cartridge into one private shelf, rebuild TypeScript and C11/WebAssembly from source, relock the anchor games, and produce one canonical private download plus one ROM-free public engine package.

Core language	C11 compiled to WebAssembly
Browser application	TypeScript, dedicated emulator worker, AudioWorklet
Host	.NET 10 MVC
Renderer ladder	WebGPU → WebGL2 → Canvas 2D
Private test shelf	38 intended entries from owner-supplied cartridge backups
Primary quality goal	Liquid authentic cadence and an evidence-based path to 1,000 playable games
Studio/remaster tooling	Explicitly deferred until emulator compatibility and cadence are mature

Contents

1	Executive Snapshot	2
1.1	Settled architecture	2
2	Non-Negotiable Project Decisions	2
3	Completed Iteration Ledger	3
3.1	Iteration 01 — Mapper-first foundation	3
3.2	Iteration 02 — MMC5 first stage and Four Score	3
3.3	Iteration 02A series — Build, renderer, UI, and audio hardening	3
3.4	Mapper and audio long-play hardening branch	4
3.5	Iteration 02B.2 — DMA and PPU edge hardening	4
3.6	Iteration 02B.3A — Controller and host-audio lifecycle	4
3.7	Kid Icarus branch — MMC1, canonical hashes, and frame transport	4
4	Validated Anchor Games	5
5	Private Cartridge Test Library	5
5.1	Existing anchors	5
5.2	Uploaded batch A — 15 archives	5
5.3	Uploaded batch B — 20 archives	6
5.4	Mandatory ingestion metadata	6
6	Current Branch Divergence and Canonical Merge	6
6.1	Three authoritative donors	7
6.2	Files that require reconciliation, not blind replacement	7
7	Immediate Planned Iterations	7
7.1	02B.4A — Canonical source merge	7
7.2	02B.4B — Complete private shelf integration	7
7.3	02B.4C — Automated mapper census and boot sweep	8
8	Future Compatibility Waves	8
8.1	Wave 1 — Common discrete mappers	8
8.2	Wave 2 — Complete MMC1/SxROM family	8
8.3	Wave 3 — MMC3, MMC6, and TQROM	8
8.4	Wave 4 — Hostile PPU timing	8
8.5	Wave 5 — Advanced Japanese mappers and expansion audio	9
8.6	Wave 6 — CPU, bus, DMA, and full PPU conformance	9
8.7	Wave 7 — Saves and deterministic state systems	9
8.8	Wave 8 — Liquid browser presentation	9
8.9	Wave 9 — Compatibility evidence ladder	9
8.10	Wave 10 — Release hardening	10
8.11	Deferred wave — Overlift Studio and remaster authoring	10
9	Build and Deployment Contract	10
10	Known Unfinished Boundaries	10
11	New-Chat Execution Starter	11
12	Final Acceptance Checklist	11

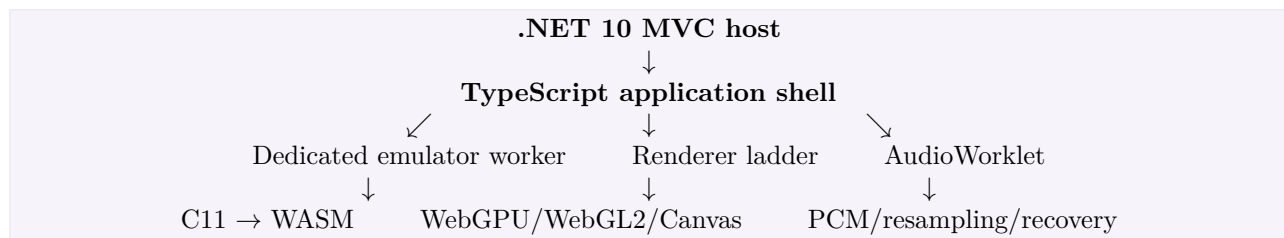
1 Executive Snapshot

The central truth

Overlift NES is no longer a speculative shell. It has three real playable anchors, a clean-room C11/WebAssembly core, modern four-controller support, a hardened AudioWorklet path, explicit DMA and PPU timing improvements, a three-tier renderer, and excellent measured Kid Icarus core headroom. The immediate engineering risk is branch divergence, not feasibility.

Current state	Three valuable branches contain non-overlapping advances and must be reconciled before further mapper expansion.
Immediate delivery	One canonical private project ZIP containing the unified source and every valid supplied game; one public ZIP containing zero ROMs.
Validated anchors	Castlevania III (MMC5), Batman: Return of the Joker (FME-7), and Kid Icarus (MMC1).
Performance evidence	Kid Icarus averaged roughly 2.02–2.05 ms per authoritative emulated frame in two matching 54,500-frame runs, providing more than eight times real-time machine-core headroom in the validation environment.
Compatibility policy	A recognized header or visible title screen is not a pass. Only documented playable checkpoints, stable audio/video/input, deterministic repeat runs, save verification where relevant, and soak evidence count.
Content policy	Commercial backups stay private and outside <code>wwwroot</code> ; public deployment and public source packages contain zero ROMs.

1.1 Settled architecture



The emulation clock is authoritative. CPU, PPU, APU, DMA, mapper state, input sampling, and deterministic hashes belong to the machine core. Presentation may drop obsolete display frames or interpolate cosmetic values, but it may never alter game timing, collision, scoring, AI, RNG, or audio pitch.

2 Non-Negotiable Project Decisions

Area	Standing decision
NES machine core	C11. Do not rewrite the NES core in Rust. WebAssembly is the browser boundary.
Browser runtime	TypeScript application and worker protocol; no JavaScript call per emulated instruction.
Host	.NET 10 MVC with a simple local <code>dotnet run</code> experience.
Build	<code>dotnet run</code> , <code>dotnet build</code> , and <code>dotnet publish</code> must compile current TypeScript and current C/WASM source offline after one-time toolchain setup.
Execution	Deterministic machine work stays in a dedicated Web Worker.
Audio	AudioWorklet delivery is independent of the visual render loop. Sound defaults to On/armed and resumes on the first valid browser gesture.
Rendering	WebGPU primary, WebGL2 GPU fallback, Canvas 2D authentic-output fallback.
Controllers	Four simultaneous modern controllers through an emulated NES Four Score; keyboard fallback remains on player one.
Compatibility	Generic cartridge-board implementations only. No game-specific hacks disguised as mapper support.

Area	Standing decision
Diagnostics	Demand-driven. Detailed streams run only while Diagnostics is open; hidden performance telemetry is shut off.
Packaging	Lean source ZIPs; exclude screenshots, caches, <code>node_modules</code> , <code>bin</code> , <code>obj</code> , and generated build output.
Public/private boundary	Private development packages may contain the owner's supplied backups. Public packages and deployment output contain zero ROMs and no <code>PrivateCartridges</code> directory.
Studio	Tile/sprite inspectors, semantic profiles, and remaster authoring remain deferred.

3 Completed Iteration Ledger

3.1 Iteration 01 — Mapper-first foundation

- Converted the supplied .NET 10 MVC project into a custom Overlift NES application.
- Removed the template `lib` directory, Bootstrap, and jQuery.
- Built a clean-room C11-to-WebAssembly core with initial 2A03 CPU, 2C02 PPU, APU, controllers, OAM DMA, and a mapper registry.
- Added Mapper 5 / MMC5 for Castlevania III and Mapper 69 / Sunsoft FME-7 for Batman.
- Reached Batman first-stage output and Castlevania III opening/title execution.
- Established strict private/public cartridge separation.

3.2 Iteration 02 — MMC5 first stage and Four Score

- Fixed MMC5 scanline/frame-state handling and reached Castlevania III's authentic first stage.
- Added four browser controller ports and real Four Score serialization on \$4016/\$4017.
- Added stable assignments for Xbox, XInput, DualShock 4, DualSense, Switch Pro, exposed Joy-Cons, 8BitDo, and generic standard-mapped USB/Bluetooth controllers.
- Added conservative PlayStation/Nintendo raw-layout fallbacks and keyboard/touch fallback on player one.
- Preserved Batman's first-stage checkpoint.

3.3 Iteration 02A series — Build, renderer, UI, and audio hardening

Revision	Completed work
02A	Detected Apple Clang installations that parse <code>-target=wasm32</code> but lack a WebAssembly backend. Prevented optional compiler diagnostics from being misread as MSBuild errors.
02A.1	Reduced the television tube to about 75% desktop width, added the WASM and WEBGPU badge, truthful runtime provenance, visible cartridge boot progress, and WebGPU black-frame validation.
02A.2–02A.4	Replaced the background-rendering shortcut with PPU fetches and shifters, fixing the global background/sprite alignment defect. Added WebGPU → WebGL2 → Canvas fallback and strict source compilation for run/build/publish.
02A.5	Split local WASM production into explicit Homebrew LLVM compile and LLD/ <code>wasm-ld</code> link phases. Added independent compiler/linker discovery and actionable logs.
02A.6	Removed generated evidence images and unnecessary TypeScript distribution files, reducing private handoffs from more than 10 MiB to roughly 2 MiB without sacrificing offline source compilation.
02A.7	Repaired macOS Bash 3.2 array/nounset behavior and improved paths containing spaces.
02A.8	Removed Bash from the <code>dotnet run</code> critical path. <code>Scripts/prepare-assets.mjs</code> became the source-build authority.

Revision	Completed work
02A.9	Moved CPU/frame/hash/FPS telemetry out of the game tube; added the retractable performance drawer, 1% and 0.1% lows, sparklines, cabinet-side modal controls, blurred pause modals, demand-driven diagnostics, sound default On, and a major base-APU/AudioWorklet sweep.

3.4 Mapper and audio long-play hardening branch

- Corrected MMC5 ExRAM modes, nametable writes, IRQ acknowledge behavior, scanline recognition, CHR register selection, and battery-backed PRG-RAM preservation.
- Corrected FME-7 IRQ clock/output gating, underflow behavior, acknowledge semantics, RAM/ROM selection, and mirroring tests.
- Added deterministic multi-checkpoint routes for Castlevania III and Batman.
- Confirmed meaningful activity on all five standard NES APU channels in the tested routes.

3.5 Iteration 02B.2 — DMA and PPU edge hardening

- Replaced immediate OAM DMA copying with an explicit 513/514-cycle state machine including dummy and parity-alignment cycles.
- Scheduled DMC sample fetches with parity-dependent three/four-cycle behavior.
- Added common DMC/OAM collision handling and explicit counters.
- Added a two-PPU-dot NMI pending window and vblank-boundary suppression/cancellation behavior.
- Preserved PPU open-bus high bits on palette PPUDATA reads.
- Improved OAMDATA reads during rendering and continued sprite evaluation beyond eight accepted sprites.
- Added 54,089-frame emulated-time soaks for Castlevania III and Batman with no CPU jam and no audio-ring overrun.
- Isolated native test temporary directories to prevent interrupted/concurrent evidence contamination.

3.6 Iteration 02B.3A — Controller and host-audio lifecycle

- Prevented a different controller from hijacking a disconnected controller's reserved port.
- Allowed a known device to reclaim its original port after reconnecting under a changed browser index.
- Kept Four Score topology stable while players three/four are disconnected but reserved.
- Replaced five independent input messages with one atomic sequenced four-port snapshot.
- Released keyboard, touch, and gamepad state before modal, hidden-tab, pagehide, and freeze pauses; repolled before resume to eliminate sticky input.
- Drained both the core PCM ring and AudioWorklet queue during pause/load/reset/resume.
- Added AudioWorklet processor, closed AudioContext, default output-device, bfcache, and page-freeze recovery.
- Added short attack/release ramps to reduce pause/resume clicks.

3.7 Kid Icarus branch — MMC1, canonical hashes, and frame transport

Numbering note

This branch was also labeled 02B.1, colliding with an earlier mapper-hardening package. In the canonical merge and future discussions, call it the **Kid Icarus branch**.

- Added generic Mapper 1 / MMC1 serial loading, bit-7 reset, RMW consecutive-write suppression, all mirroring modes, 32 KiB/16 KiB PRG modes, 8 KiB/split 4 KiB CHR modes, CHR-RAM writes, and PRG-RAM protection.
- Added SNROM behavior plus initial SOROM, SUROM, and SXROM banking groundwork.
- Permanently added Kid Icarus and reached its first playable vertical chamber with movement, jumping, and firing.
- Replaced raw-structure state hashing with canonical field-by-field little-endian hashing and exposed CPU, PPU, APU, mapper, and whole-machine hashes.
- Added a bounded three/four-buffer transferable framebuffer pool, superseded-frame recycling, stale-presentation dropping without skipping machine work, and high-refresh duplicate GPU-submission suppression.
- Fixed immediate one-time redraw after presentation-mode changes.

4 Validated Anchor Games

Game	Mapper	Frame	Locked evidence
Castlevania III: Dracula's Curse	5 / MMC5	3,000	Framebuffer 8ab13b1e20a49e7c; 02B.2 state C4ABD5F8; PCM 2,395,258 / b0fe4f73e9773744.
Batman: Return of the Joker	69 / FME-7	1,000	Framebuffer 39d0f1bd48f30942; 02B.2 state C2743BEC; PCM 797,889 / 8c68dcf178d10138.
Kid Icarus: Angel Land Story	1 / MMC1	1,500	Framebuffer d99ebd822b0819bc; canonical state 912ACFE3; PCM 1,197,231 / e01b0358d171c28f.

Relock after merge

These state hashes are branch-specific. The unified 02B.4A core may legitimately change state hashes because ABI layout, mapper state, DMA state, or timing state changes. Preserve the known visual/audio behavior, then generate and document new canonical locks from the merged tree.

5 Private Cartridge Test Library

The intended private shelf contains **38 entries**: three existing anchors and 35 newly supplied archives. All were already uploaded. The next chat must retrieve them from the conversation/Library and must not ask Sean to upload them again.

5.1 Existing anchors

1. Batman: Return of the Joker
2. Castlevania III: Dracula's Curse
3. Kid Icarus: Angel Land Story

5.2 Uploaded batch A — 15 archives

1. Megami Tensei — English MMC3 Patch
2. Lagrange Point
3. Mouryou Senki Madara
4. Akumajou Densetsu
5. Gimmick!
6. Micro Machines
7. Marble Madness
8. Battletoads
9. Pin-Bot
10. StarTropics

11. Kirby's Adventure Rev A
12. Deadly Towers
13. Gumshoe

14. IronSword: Wizards & Warriors II
15. Solstice

5.3 Uploaded batch B — 20 archives

1. Donkey Kong Classics
2. Super Mario Bros. / Duck Hunt
3. Final Fantasy I & II
4. Metroid
5. Zelda II: The Adventure of Link
6. A Boy and His Blob
7. Nobunaga's Ambition II
8. Nobunaga's Ambition
9. Romance of the Three Kingdoms
10. Dragon Warrior IV

11. Dragon Warrior III
12. Dragon Warrior II
13. Dragon Warrior
14. Teenage Mutant Ninja Turtles
15. Mike Tyson's Punch-Out!!
16. Contra
17. Super Mario Bros. 2
18. Super Mario Bros. 3
19. The Legend of Zelda
20. Dr. Mario Rev 1

5.4 Mandatory ingestion metadata

For every valid candidate, generate and lock:

- Display title and exact filename
- SHA-256 and CRC-32
- iNES or NES 2.0 format
- Mapper and submapper
- PRG-ROM and CHR-ROM sizes
- CHR-RAM and PRG-RAM requirements
- Battery/save type
- Trainer presence
- Mirroring and region
- Reset vector
- Current compatibility classification

Megami Tensei special rule

If the archive contains a complete patched `.nes` image, catalog it as a distinct modified build. If it contains only IPS/BPS/UPS patch data, quarantine it as **Base cartridge required**. Do not silently manufacture or distribute an unsupplied base image.

6 Current Branch Divergence and Canonical Merge

6.1 Three authoritative donors

Source package	Authority in the merge
02B.3A Controller/Audio Lifecycle	Application/client base, worker lifecycle, atomic controller transport, controller reservation, sticky-input protection, host/core audio recovery.
02B.2 DMA/PPU Edge Hardening	Machine-core timing donor: explicit OAM DMA, scheduled DMC DMA, collision handling, NMI pending/suppression, open bus, OAMDATA, sprite evaluation.
Kid Icarus branch	MMC1/SxROM, Kid Icarus fixture/replay, canonical subsystem hashing, bounded frame transport, high-refresh duplicate-submission suppression.

6.2 Files that require reconciliation, not blind replacement

- | | |
|-----------------------------------|---------------------------------------|
| • Core/C/nes_core.c | • Scripts/prepare-assets.mjs |
| • Core/C/nes_core.h | • Tests/* |
| • Client/src/main.ts | • PrivateCartridges/catalog.json |
| • Client/src/emu.worker.ts | • nesgpu.csproj |
| • Client/src/protocol.ts | • wwwroot/js/* |
| • Client/src/controller-bridge.ts | • wwwroot/wasm/overlift-nes-core.wasm |
| • Client/src/audio.worklet.ts | • wwwroot/build/runtime-manifest.json |
| • Client/src/renderers/* | |

7 Immediate Planned Iterations

7.1 02B.4A — Canonical source merge

Objective: one source tree containing the strongest machine core, browser lifecycle, MMC1 work, canonical hashes, and frame transport.

Exit gates:

- Core ABI reconciled and documented.
- Strict TypeScript build passes.
- C11 warning-clean native build passes.
- Fresh wasm32 object and `wasm-ld` module pass validation.
- Batman, Castlevania III, and Kid Icarus regressions pass from power-on.
- Controller/Four Score, controller lifecycle, AudioWorklet lifecycle, DMA/PPU, mapper, renderer-policy, and hidden-tab tests pass.
- `dotnet run` remains the one-command offline owner workflow.

7.2 02B.4B — Complete private shelf integration

Objective: safely extract every valid supplied image into `PrivateCartridges`, generate the exact catalog, and make every valid title visible in the Cartridges modal.

Exit gates:

- Path traversal, nested executable, malformed-header, and duplicate payloads rejected.

- Patch-only inputs quarantined honestly.
- Unsupported titles remain visible with **Mapper recognized; implementation pending** rather than disappearing.
- Private ZIP contains the full validated shelf; public ZIP contains zero ROMs and no **PrivateCartridges** directory.
- Both final ZIPs are clean-extracted, checksum-verified, and their exact ROM counts reported.

7.3 02B.4C — Automated mapper census and boot sweep

Every valid image receives a common initial harness:

Load → reset → 60 frames → 600 frames → deterministic Start → 1,800 frames → classify

Capture CPU jam, run-state faults, visible color count, framebuffer/state/audio hashes, channel activity, mapper identity, and first evidence of accepted input. A title screen is *not* a playable classification.

8 Future Compatibility Waves

8.1 Wave 1 — Common discrete mappers

Principal games: Super Mario Bros./Duck Hunt, Contra, Donkey Kong Classics, Solstice, IronSword, Deadly Towers, Gumshoe.

Hardware targets: NROM, UxROM/UNROM/UOROM, CNROM, AxROM, BNROM, GxROM, Camerica variants, one-screen mirroring, and bus-conflict behavior.

Gate: generic mapper self-tests, native/WASM agreement, and at least one real playable checkpoint per board family without title-specific branches.

8.2 Wave 2 — Complete MMC1/SxROM family

Principal games: Kid Icarus, The Legend of Zelda, Zelda II, Metroid, A Boy and His Blob, TMNT, Dragon Warrior I–IV, Nobunaga’s Ambition I–II, Romance of the Three Kingdoms, Final Fantasy I & II, Dr. Mario.

Hardware targets: SEROM/SHROM, SNROM, SOROM, SUROM, SXROM, MMC1A/MMC1B distinctions, banked PRG-RAM, outer PRG banking, battery persistence, reset/power-cycle retention.

Kid Icarus butter gate: doors, shops, item interactions, death/restart, password flow, vertical progression, fortress rooms, boss route, one-hour deterministic run, and physical browser A/V/controller acceptance.

8.3 Wave 3 — MMC3, MMC6, and TQROM

Principal games: Super Mario Bros. 2, Super Mario Bros. 3, Kirby’s Adventure, StarTropics, Pin-Bot, and a runnable Megami Tensei patched image.

Hardware targets: qualified PPU A12 edge tracking, IRQ reload/acknowledge edges, PRG/CHR inversion, MMC6 RAM protection, TQROM CHR-ROM/CHR-RAM selection, status-bar splits, dense bank switching.

Gate: multiple scrolling gameplay checkpoints, stable IRQ splits with no status shake, correct saves, and deterministic 15-minute runs.

8.4 Wave 4 — Hostile PPU timing

Principal games: Battletoads, Marble Madness, Micro Machines, Mike Tyson’s Punch-Out!!

Targets: sprite-zero timing, 34th background-tile fetch behavior, mid-scanline CHR switching, rendering enable/disable edges, OAM reads during rendering, diagonal sprite-overflow bug, NMI/PPUDATA race cases, CPU/PPU phase alignment, CHR-RAM streaming.

8.5 Wave 5 — Advanced Japanese mappers and expansion audio

Principal games: Akumajou Densetsu, Mouryou Senki Madara, Lagrange Point, Gimmick!, plus deeper Batman and Castlevania III routes.

Targets: VRC6a/VRC6b wiring, VRC6 pulse/saw, VRC7 six-channel FM, Sunsoft 5B synthesis, MMC5 pulse/PCM conformance, base/expansion mixing, mapper IRQs, per-channel activity and PCM hashes.

8.6 Wave 6 — CPU, bus, DMA, and full PPU conformance

- Exact DMC conflicts with controller, PPU, and CPU reads.
- DMC/OAM overlap at all alignments and instruction bus ordering.
- Interrupt edge behavior and remaining CPU timing fixtures.
- Complete PPU register-race family, exact sprite overflow, odd-frame timing, mapper-visible PPU address timing.
- NTSC, PAL, and Dendy timing modes.

8.7 Wave 7 — Saves and deterministic state systems

Implement battery persistence, versioned save states, recorded input replay, rewind checkpoints, crash-safe IndexedDB/OPFS writes, state migration, and deterministic restore. The same state plus the same input must reproduce the same CPU, PPU, APU, mapper, framebuffer, and PCM hashes.

8.8 Wave 8 — Liquid browser presentation

- SharedArrayBuffer fast path when cross-origin isolated; transferable-buffer fallback everywhere else.
- Worker-owned deadlines and bounded catch-up behavior.
- No steady-state framebuffer allocations, unbounded queues, duplicate high-refresh uploads, or growing input latency.
- Audio remains independent of visual cadence.
- Stable 60/90/120/144 Hz and variable-refresh presentation.
- Tab suspension, AudioContext loss, output-device change, controller reconnect, and GPU-device-loss recovery.
- Thermal/battery-aware presentation policy that never lowers machine accuracy.

Authentic-player acceptance targets:

Average presentation FPS	≥ 59.5
1% low	≥ 58 FPS
0.1% low	≥ 55 FPS
Recurring main-thread long tasks	0
Sustained core audio underruns/overruns	0
Unbounded frame/input/audio queue growth	0

8.9 Wave 9 — Compatibility evidence ladder

25 deeply tested → 100 → 250 → 500 → 1,000 Playable

A title counts as **Playable** only after verified identity, correct board classification, accepted input, a genuine gameplay checkpoint, acceptable graphics/audio, save verification where relevant, a deterministic repeat run, at least a 15-minute stability result, and no unresolved critical defect.

8.10 Wave 10 — Release hardening

Cross-browser and cross-OS matrix; integrated-GPU and phone tiers; physical Xbox/PlayStation/Switch/Joy-Con/8BitDo sessions; eight-hour unattended soak; GPU/audio/controller recovery; source provenance; third-party notices; rights-cleared public content; and a production package with zero commercial ROMs.

8.11 Deferred wave — Overlift Studio and remaster authoring

Do not begin tile/sprite inspectors, click-to-trace, semantic event authoring, replacement meshes, voxel profiles, or profile-pack tooling until the canonical emulator is unified, save systems are stable, cadence is liquid, and the compatibility matrix is substantial.

9 Build and Deployment Contract

The owner toolchain is:

- .NET 10 SDK
- Node.js
- Homebrew LLVM with wasm32 backend
- Homebrew LLD / `wasm-ld`
- Bundled TypeScript 5.8.3 compiler subset

Normal operation remains:

```
cd nesgpu && dotnet run
```

Expected chain:

TypeScript source → browser JavaScript

C11 source → wasm32 object → `wasm-ld` → validated WASM

C#/Razor → .NET 10 build → Kestrel launch

The source-build critical path must not depend on Bash. Compiler and linker arguments must remain safe for paths containing spaces. `dotnet publish -c Release` must invoke the same fresh asset build and fail when private content enters publish output.

10 Known Unfinished Boundaries

Do not claim completion of the following:

- Exact bus microcycles for every CPU access
- PAL and Dendy timing
- Every DMC side-effecting read conflict
- Battery persistence
- Exact diagonal sprite overflow
- Save states, replay, and rewind
- Full PPU register-race family
- Famicom Disk System
- Complete MMC1 board matrix
- Physical controller acceptance matrix
- MMC3/MMC6/TQROM
- One unified 38-entry private package
- Most discrete mapper families
- A 1,000-playable-game evidence claim
- VRC6/VRC7 and Sunsoft 5B audio
- Studio/remaster authoring
- Complete MMC5 expansion audio

11 New-Chat Execution Starter

Paste the following into the new chat. It is intentionally operational rather than conversational.

Continue Overlift NES from the Master Engineering Handoff.

This is a code, test, merge, and packaging task, not another planning response.

Retrieve these existing source packages from my Library:

1. Overlift.NES.Iteration-02B.3A-Controller-Audio-Lifecycle-Private.zip
2. Overlift.NES.Iteration-02B.2-DMA-PPU-Edge-Hardening-Private.zip
3. Overlift.NES.Iteration-02B.1-MMC1-Kid-Icarus-Frame-Pacing-Private.zip

Retrieve every NES cartridge upload from both July 26 batches, plus Batman, Castlevania III, and Kid Icarus. Do not ask me to upload them again.

Perform Iteration 02B.4A and 02B.4B.

Use 02B.3A as the application/client base. Merge the 02B.2 DMA/PPU core. Merge the Kid Icarus branch's MMC1, canonical hashes, bounded framebuffer transport, high-refresh submission suppression, fixture, and tests.

Do not blindly replace diverged files. Reconcile the C core/header, worker protocol, TypeScript client, renderer transport, AudioWorklet lifecycle, tests, project configuration, package scripts, and runtime manifests.

Safely extract every valid .nes image into PrivateCartridges. Calculate SHA-256, CRC-32, header format, mapper, submapper, PRG/CHR sizes, RAM, battery, trainer, mirroring, region, and reset vector. Reject malformed, duplicate, traversal, nested executable, and patch-only payloads.

Treat Megami Tensei - English MMC3 Patch honestly: include only a complete runnable .nes image; otherwise label it Base cartridge required.

Generate the private cartridge catalog automatically. Show every valid entry in the Cartridges modal, including unsupported boards with an honest status.

Recompile TypeScript and C11 to fresh WebAssembly. Rerun and relink Batman, Castlevania III, Kid Icarus, controller/Four Score, controller lifecycle, AudioWorklet lifecycle, DMA/PPU, mapper, and renderer tests.

Preserve one-command offline build through: dotnet run

Produce:

1. One canonical private source ZIP with all current work and valid cartridges.
2. One public engine ZIP containing zero ROMs and no PrivateCartridges folder.
3. SHA-256 files for both ZIPs.
4. A complete 02B.4A/02B.4B validation report.
5. A generated cartridge inventory and initial mapper census.

Studio, tile inspection, semantic authoring, and remaster work remain deferred. Do not report completion until the final private ZIP has been clean-extracted, its checksum manifest verified, and its exact ROM count confirmed.

12 Final Acceptance Checklist

- | | |
|---|--|
| ☐ Three branches retrieved | ☐ Batman regression passed |
| ☐ All cartridge archives retrieved | ☐ Castlevania III regression passed |
| ☐ Canonical merge completed | ☐ Kid Icarus regression passed |
| ☐ C core warning-clean | ☐ Controller lifecycle passed |
| ☐ TypeScript strict build passed | ☐ Audio lifecycle passed |
| ☐ Fresh WASM linked and validated | ☐ DMA/PPU tests passed |
| ☐ Core ABI reconciled | ☐ Full private shelf generated |

- | | |
|--|---|
| ☐ Patch-only inputs quarantined | ☐ Both ZIPs clean-extracted |
| ☐ Private ZIP generated | ☐ Checksums verified |
| ☐ Public ROM-free ZIP generated | ☐ Validation report attached |

Bottom line

The emulator has already crossed the feasibility threshold. The next disciplined move is consolidation: merge the strongest three lines, integrate the full private test library, relock the three anchor games, and then advance mapper wave by mapper wave without sacrificing cadence, audio stability, or honest compatibility evidence.

Source Provenance for This Handoff

This document consolidates the completed Overlift NES validation reports, branch package records, owner-supplied cartridge inventory, and the supplied Overlift Studio technical design. The design principle remains: mature cores compile to WebAssembly, emulation timing is authoritative, the renderer is a consumer, and WebGPU ships with a real WebGL2/original fallback. Commercial cartridge data remains private; public packages remain engine-only and rights-clean.